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ST4003CEM Object Oriented Programming

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1. Write a program to check whether a person can cast a vote or not. The condition is you must be over 18 years old to vote.

IsVoter.java > IsVoter

```java
import java.util.Scanner;

public class IsVoter {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner=new Scanner(System.in);
        System.out.print(s: "Enter name:");
        String name=scanner.nextLine();
        System.out.print(s: "Enter age:");
        int age=scanner.nextInt();
        if (age>=18){
            System.out.printf(format: "Hello %s, you can vote.",name);
        }
        else{
            System.out.printf(format: "Hello %s, you can't vote.",name);
        }

        scanner.close();
    }
}
```

OUTPUT DEBUG CONSOLE PROBLEMS TERMINAL EXPLORER EXPLORER

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\Users\sthan\OneDrive\Desktop
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sthan\AppData\Roaming\
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'IsVoter'
Enter name:Nischal
Enter age:18
Hello Nischal, you can vote.
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>
```

2. Write a program to calculate Simple Interest.
   Formula Simple Interest = (PrincipleAmount*Time*Rate/100);

SimpleIntrest.java > SimpleIntrest > main(String[])

```java
import java.util.Scanner;
public class SimpleIntrest {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner=new Scanner(System.in);
        System.out.print(s: "Enter Principle amount: ");
        float PrincipleAmount=scanner.nextFloat();
        System.out.print(s: "Enter Rate: ");
        float Rate=scanner.nextFloat();
        System.out.print(s: "Enter Time in year: ");
        float Time=scanner.nextFloat();
        float SI=(PrincipleAmount*Time*Rate)/100;
        System.out.println("Simple intrest:"+SI);
        scanner.close();
    }
}
```

OUTPUT DEBUG CONSOLE PROBLEMS TERMINAL EXPLORER EXPLORER

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  & 'C:\Program Files\Java\jdk-26\bin\j
' 'C:\Users\sthan\AppData\Roaming\Code\User\workspaceStorage\ed72d0180b42b81006fdab86cb95ded9\redhat.java
Enter Principle amount: 1000
Enter Rate: 12
Enter Time in year: 2
Simple intrest:240.0
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>
```

3. Write a program to calculate, area of the triangle, and the Volume of the Cube and Cuboid.

   **Formula: Volume of Cuboid = length*width*height; and Volume of Cube: side*side*side;**

CommandLineExplain.java SwitchCase.java Grade

AreaVolume.java > AreaVolume > main(String[])

```java
import java.util.Scanner;
public class AreaVolume {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner=new Scanner(System.in);
        System.out.println(x: "1: Area to Triangle \n2: Volume of Cube \n3: Volume of Cuboid \nChooise:");
        int chooise=scanner.nextInt();
        switch (chooise) {
            case 1:
                System.out.println(x: "Area of Triangle");
                System.out.print(s: "Enter Base:");
                float base=scanner.nextFloat();
                System.out.print(s: "Enter Height:");
                float height=scanner.nextFloat();
                float area=(base*height)/2;
                System.out.println("Area:"+area);
                break;
            case 2:
                System.out.println(x: "Volume of Cube");
                System.out.print(s: "Enter side:");
                float side=scanner.nextFloat();
                float volume_cube=side*side*side;
                System.out.println("Volume:"+volume_cube);

                break;
            case 3:
                System.out.println(x: "Volume of Cuboid");
                System.out.print(s: "Enter Length:");
                float Length=scanner.nextFloat();
                System.out.print(s: "Enter Width:");
                float Width=scanner.nextFloat();
                System.out.print(s: "Enter Height:");
                float Height=scanner.nextFloat();
                float volume_cuboid=Length*Width*Height;
                System.out.println("Volume:"+volume_cuboid);
                break;
            default:
                System.out.println(x: "Invalid Chooise");
                break;
            }
            scanner.close();

    }
}
```

```
OUTPUT    DEBUG CONSOLE    PROBLEM

PS C:\Users\sthan\OneDrive\Desk
\Java\jdk-26\bin\java.exe' '-X
95ded9\redhat.java\jdt_ws\java
1: Area to Triangle
2: Volume of Cube
3: Volume of Cuboid
Chooise:
1
Area of Triangle
Enter Base:4
Enter Height:10
Area:20.0
PS C:\Users\sthan\OneDrive\Desk
\Java\jdk-26\bin\java.exe' '-X
95ded9\redhat.java\jdt_ws\java
1: Area to Triangle
2: Volume of Cube
3: Volume of Cuboid
Chooise:
2
Volume of Cube
Enter side:10
Volume:1000.0
PS C:\Users\sthan\OneDrive\Desk
\Java\jdk-26\bin\java.exe' '-X
95ded9\redhat.java\jdt_ws\java
1: Area to Triangle
2: Volume of Cube
3: Volume of Cuboid
Chooise:
3
Volume of Cuboid
Enter Length:2
Enter Width:5
Enter Height:10
Volume:100.0
PS C:\Users\sthan\OneDrive\Desk
```

4. Write the ternary operator for question no. 1

IsVoter.java > ...

```java
import java.util.Scanner;

public class IsVoter {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner=new Scanner(System.in);
        System.out.print(s: "Enter name:");
        String name=scanner.nextLine();
        System.out.print(s: "Enter age:");
        int age=scanner.nextInt();
        String voter=(age>=18)?"Hello "+name+", you can vote.":"Hello"+name+",you can't vote.";
        System.out.println(voter);
        scanner.close();
    }

}
```

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp'
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'IsVoter'
Enter name:Nischal
Enter age:18
Hello Nischal, you can vote.
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp'
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'IsVoter'
Enter name:Nischal 13
Enter age:13
HelloNischal 13,you can't vote.
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>
```

5. Write a program to take two integer inputs from the user and print the sum and product of them.

SumAndProduct.java > SumAndProduct > main(String[])

```java
import java.util.Scanner;
public class SumAndProduct {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner=new Scanner(System.in);
        System.out.print(s: "Enter first number:");
        int num1=scanner.nextInt();
        System.out.print(s: "Enter second number:");
        int num2=scanner.nextInt();
        int sum=num1+num2;
        int product =num1*num2;
        System.out.println("Sum is "+sum);
        System.out.println("Product is "+product);
        scanner.close();
    }
}
```

OUTPUT DEBUG CONSOLE PROBLEMS TERMINAL EXPLORER EXPLORER

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  & 'C:\Pro
' 'C:\Users\sthan\AppData\Roaming\Code\User\workspaceStorage\ed72d0180b42b810
Enter first number:12
Enter second number:11
Sum is 23
Product is 132
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>
```

6. Take two integer inputs from the user. First, calculate the sum of two, then the product of two. Finally, calculate the division of the thus obtained sum and product and print the result.

SumAndProduct.java > SumAndProduct

```java
import java.util.Scanner;
public class SumAndProduct {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner=new Scanner(System.in);
        System.out.print(s: "Enter first number:");
        int num1=scanner.nextInt();
        System.out.print(s: "Enter second number:");
        int num2=scanner.nextInt();
        int sum=num1+num2;
        int product =num1*num2;
        System.out.println("Sum is "+sum);
        System.out.println("Product is "+product);
        scanner.close();
        float division=(float)sum/product;
        System.out.println("Division of sum and product is "+division);
    }
}
```

OUTPUT DEBUG CONSOLE PROBLEMS TERMINAL EXPLORER EXPLORER

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\Users\sthan\OneDrive\Deskt
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sthan\AppData\Roamin
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'SumAndProduct'
Enter first number:10
Enter second number:5
Sum is 15
Product is 50
Division of sum and product is 0.3
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>
```

7. Take the name, roll number, and field of interest from the user and print in the format below: Hey, my name is XYZ and my roll number is XYZ. My field of interest are xyz.

UserDataFormate.java > UserDataFormate > main(String[])

```java
import java.util.Scanner;
public class UserDataFormate {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner=new Scanner(System.in);
        System.out.print(s: "Enter name:");
        String name=scanner.nextLine();
        System.out.print(s: "Enter role number:");
        int rollNumber=scanner.nextInt();
        scanner.nextLine();
        System.out.print(s: "Enter field of interest:");
        String fieldOfIntrest=scanner.nextLine();

        System.out.printf(format: "Hey, my name is %s and my role number is %d. My field of intrest is %s",name,rollNumb
        scanner.close();

    }
}
```

OUTPUT DEBUG CONSOLE PROBLEMS TERMINAL EXPLORER EXPLORER

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd 'c:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz'; & 'C:\Program Files'
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sthan\AppData\Roaming\Code\User\workspaceStorage\ed72d0180b42b81006fdab86cb
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'UserDataFormate'
Enter name:Nischal Shrestha
Enter role number:250412
Enter field of interest:Vibe coding
Hey, my name is Nischal Shrestha and my role number is 250412. My field of intrest is Vibe coding
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>
```

8. Take side of a square from user and print area and perimeter of it. Also, calculate Simple Interest, Area of triangle and Volume of Cube and Cuboid. Take the attributes as user input.

GeometryAndInterest.java > GeometryAndInterest >

```java
import java.util.Scanner;
public class GeometryAndInterest {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        //  Square
        System.out.print(s: "Enter side of square: ");
        float side = scanner.nextFloat();
        float squareArea = side * side;
        float squarePerimeter = 4 * side;
        // Simple Interest
        System.out.print(s: "Enter Principal: ");
        float p = scanner.nextFloat();
        System.out.print(s: "Enter Rate: ");
        float r = scanner.nextFloat();
        System.out.print(s: "Enter Time: ");
        float t = scanner.nextFloat();
        float si = (p * r * t) / 100;
        // Triangle
        System.out.print(s: "Enter base of triangle: ");
        float base = scanner.nextFloat();
        System.out.print(s: "Enter height of triangle: ");
        float height = scanner.nextFloat();
        float triangleArea = (base * height) / 2;
        //  Cube
        System.out.print(s: "Enter side of cube: ");
        float cubeSide = scanner.nextFloat();
        float cubeVolume = cubeSide * cubeSide * cubeSide;
        // Cuboid
        System.out.print(s: "Enter length of cuboid: ");
        float length = scanner.nextFloat();
        System.out.print(s: "Enter breadth of cuboid: ");
        float breadth = scanner.nextFloat();
        System.out.print(s: "Enter height of cuboid: ");
        float heightCuboid = scanner.nextFloat();
        float cuboidVolume = length * breadth * heightCuboid;
        // Output
        System.out.println(x: "\n--- Results ---");
        System.out.println("Square Area: " + squareArea);
        System.out.println("Square Perimeter: " + squarePerimeter);
        System.out.println("Simple Interest: " + si);
        System.out.println("Triangle Area: " + triangleArea);
        System.out.println("Cube Volume: " + cubeVolume);
        System.out.println("Cuboid Volume: " + cuboidVolume);
        scanner.close();
    }
}
```

```
gram Files\Java\jdk-26\bin\java.exe' '-XX:+ShowCod
180b42b81006fdab86cb95ded9\redhat.java\jdt_ws\java
Enter side of square: 4
Enter Principal: 400
Enter Rate: 12
Enter Time: 2
Enter base of triangle: 5
Enter height of triangle: 10
Enter side of cube: 3
Enter length of cuboid: 5
Enter breadth of cuboid: 4
Enter height of cuboid: 3

--- Results ---
Square Area: 16.0
Square Perimeter: 16.0
Simple Interest: 96.0
Triangle Area: 25.0
Cube Volume: 27.0
Cuboid Volume: 60.0
PS C:\Users\sthan\OneDrive\Desktop\Softwarica seco
```

9. Ask user to give two double input for length and breadth of a rectangle and print area type casted to int.

TypeCasting.java > TypeCasting > main(String[])

```java
import java.util.Scanner;
public class TypeCasting {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner=new Scanner(System.in);
        System.out.print(s: "Enter Length:");
        float length=scanner.nextFloat();
        System.out.print(s: "Enter breadth:");
        float breadth= scanner.nextFloat();
        int area= (int)(length*breadth);
        System.out.println("Area of Rectriangle:"+area);

        scanner.close();
    }
}
```

OUTPUT DEBUG CONSOLE PROBLEMS TERMINAL EXPLORER EXPLORER

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  & 'C:\P
' 'C:\Users\sthan\AppData\Roaming\Code\User\workspaceStorage\ed72d0180b42b8
Enter Length:50
Enter breadth:20
Area1000
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>  c:; cd
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp'
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin' 'TypeCasting'
Enter Length:50
Enter breadth:20
Area of Rectriangle:1000
PS C:\Users\sthan\OneDrive\Desktop\Softwarica second sem\java clz>
```

10. Write a program to calculate the total marks of four subjects of a student and the total percentage secured. And use the following conditions to generate the final result;
    a. If equal to or more than 70 -> First Class
    b. If more than 59 -> Upper Second Class
    c. If more than 49 -> Second class
    d. If more than 39 -> Third class and if below than 40 the result is fail.

MarksCalculator.java > ...

```java
import java.util.Scanner;
public class MarksCalculator {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner=new Scanner(System.in);
        System.out.println(x: "Enter marks of 4 subjects out of 100");
        float m1=scanner.nextFloat();
        float m2=scanner.nextFloat();
        float m3=scanner.nextFloat();
        float m4=scanner.nextFloat();
        float total=m1+m2+m3+m4;
        float percentage=total/4;
        if (percentage>=70){
            System.out.println("First Class and total:"+total+" percentage:"+percentage);
        }
        else if (percentage>59){
            System.out.println("Upper Second Class and total:"+total+" percentage:"+percentage);
        }
        else if (percentage>49){
            System.out.println("Second Class and total:"+total+" percentage:"+percentage);
        }
        else if (percentage>39){
            System.out.println("Third Class and total:"+total+" percentage:"+percentage);
        }
        else if (percentage<40){
            System.out.println(x: "Fail");
        }
        else{
            System.out.println(x: "Somewant wrong");
        }
        scanner.close();
    }
}
```

```
PS C:\Users\sthan\OneDrive\Desktop\Softwarica se
\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetails
95ded9\redhat.java\jdt_ws\java clz_18bc6240\bin'
Enter marks of 4 subjects out of 100
70
77
75
79
First Class and total:301.0 percentage:75.25
PS C:\Users\sthan\OneDrive\Desktop\Softwarica se
```